Case Studies in Thermal Engineering 22 (2020) 100754

Contents lists available at ScienceDirect

Case Studies in Thermal Engineering

journal homepage: http://www.elsevier.com/locate/csite

Dynamic thermal modelling of PV performance and effect of heat capacity on the module temperature

B. Tuncel [a,c,f,*], T. Ozden [b,c], R.S. Balog [c,d], B.G. Akinoglu [a,c,e]

[a] Department of Physics, Faculty of Arts and Sciences, Middle East Technical University, Ankara, Turkey
[b] Department of Electrical & Electronics Engineering, Gumushane University, Gumushane, Turkey
[c] GÜNAM – Center for Solar Energy Research and Applications, Middle East Technical University, Ankara, Turkey
[d] Renewable Energy and Advanced Power Electronics Research Laboratory, Department of Electrical & Computer Engineering, Texas A&M University, College Station, USA
[e] Earth System Science Program, Graduate School of Natural and Applied Sciences, Middle East Technical University, Ankara, Turkey
[f] Department of Energy Engineering, Faculty of Engineering, Ankara University, Ankara, Turkey

ARTICLE INFO

Keywords:
Photovoltaic modules
Thermal modelling
Module temperature
Heat capacity

ABSTRACT

One of the main weaknesses of Si-based Photovoltaic (PV) solar modules is the sensitivity of their efficiency to module (cell) temperature. Especially in locations with long hot seasons, the efficiency loss of PVs due to high temperatures should be considered carefully. Thermal modelling is a method to predict the performance of a PV module using essentially the 1st law of thermodynamics and available data. In this paper, a transient thermal model is described, which considers hourly meteorological data, including wind speed and direction, module parameters, and locational information. In transient analysis, the heat capacity value of PV panel is required, but it is not a parameter specified in the manufacturer's datasheet. Experiments on the heat capacity of PV modules are missing in the literature. The thermal model provides verified performance analysis for a poly-c-Si PV module installed in Ankara, Turkey, with a calculated heat capacity value. And a sensitivity analysis for the heat capacity of the transient thermal model is performed. It is found that the model results are almost invariant under changing module heat capacity values.

1. Introduction

After the first introduction of 6% efficient Si solar cell in 1954 [1], with more than 60 years of research history, Si solar cells reached above 26% efficiency today [2]. Si has the largest share in the PV market: 95% of produced PVs in 2017 were c-Si [3]. It is predicted that Si will continue to dominate the PV market for a foreseeable future. Despite being one of the most established technologies in the PV sector, Si PV has several weaknesses. One of them is its temperature sensitivity. The efficiency of Si solar cells is severely reduced with elevated cell temperatures [4]. Therefore, the temperature of PV solar cells or modules should be considered carefully in outdoor performance analyses, especially in locations with long hot seasons and high insolation like southern and central parts of Turkey [5,6]. Various approaches are used for PV cell/module temperature predictions in literature. Extensive reviews can be found in Refs. [7,8].

In this study, a transient thermal model for PV module temperature prediction is described. The results verified with experimental measurements are presented. Eliminating location-dependent coefficients, the model is universal, and no data set is required to "train"

* Corresponding author. Department of Physics, Faculty of Arts and Sciences, Middle East Technical University, Ankara, Turkey.
E-mail address: btuncel@ankara.edu.tr (B. Tuncel).

https://doi.org/10.1016/j.csite.2020.100754
Received 12 May 2020; Received in revised form 12 August 2020; Accepted 20 September 2020
Available online 24 September 2020

the model. The model is first introduced in the paper by Balog [9] and modified and extended in the study by Tuncel [10]. The model considers meteorological data, including wind direction, which is often neglected in similar studies, module parameters, and locational information.

Heat capacity is one of the PV module parameters and a necessary part of the transient thermal analyses. However, there are no studies on the measurement of the heat capacity of PV modules in literature. The transient thermal models developed for PV modules, most of the time, consider a fixed value for heat capacity [11–14]. In the study by Torres Lobera [12], the effect of heat capacity value choice on a dynamic thermal model is investigated using measurements with 1-s intervals taken from a research plant in Finland, where cold and rainy weather dominates most of the year. The authors parametrize the heat capacity value in the model and consider the resulting deviation of the module temperature from the measured value as the model performance indicator. They conclude that the assumed value of heat capacity in the model influences the performance of the model in the summertime when the ambient temperatures are relatively higher.

The focus of this paper is to determine the effect of the heat capacity value of the PV module on the performance of the developed transient thermal model for a semi-arid continental climate with hot and dry summers. The model considers hourly input data, which are more commonly available for a location. A sensitivity analysis is carried out using a range of heat capacity values to assess the effect.

2. Thermal modelling

The basic principle in thermal modelling is the first law of thermodynamics, which is energy conservation. The PV module without the mounting frame is considered as control volume, and lumped transient analysis is adopted for this study. In the case of lumped transient thermal modelling, the governing equation derived from the first law of thermodynamics for a PV module is:

$$\frac{\mathrm{d}T_m}{\mathrm{d}t}C_m = Q_{solar} - W_{electrical} - Q_{loss} \tag{1}$$

where T_m and C_m are temperature and heat capacity of the module, Q_{solar} is the in-plane solar irradiance, $W_{electrical}$ is the electrical power produced by the module, and Q_{loss} is the rate of energy loss to the environment via heat transfer.

2.1. Heat transfer mechanisms

Heat exchange with the environment occurs with three modes of heat transfer, which are conductive, convective, and radiative heat transfer.

2.1.1. Conductive heat transfer

Conductive heat transfer occurs when there is a temperature gradient within a solid. The thickness of laminated PV cells is usually very small, around 3 mm [15]. Therefore, it is assumed that there is no significant temperature gradient within the PV module; and conduction is neglected.

2.1.2. Convective heat transfer

Convective heat transfer occurs when there is a fluid domain in contact with a heated or cooled surface. Convection is a combination of heat transfer by conduction through the fluid molecules and bulk motion of the fluid, in which heat is carried away from the surface. Convective heat transfer can be expressed simply with Newton's law of cooling:

$$Q_{conv} = h(T_m - T_\infty) \tag{2}$$

where Q_{conv} is the rate of convective heat loss, T_m and T_∞ are temperatures of module and fluid, respectively, and h is the convective heat transfer coefficient. Convective heat transfer coefficient h is calculated using dimensionless Nusselt Number.

According to the driving force behind fluid motion, two types of convection can be defined: natural convection and forced convection. Natural convection is driven by buoyancy caused by the temperature gradient within the fluid, while in forced convection, the fluid motion is provided by external forces.

In the determination of Nusselt number and consequently how much heat is transferred from a surface, fluid flow characteristics near the surface play a significant role. In general, two distinct characterizations can be observed: laminar flow, where the flow is well defined, and analytical solutions are possible, and turbulent flow, where the fluid motion is chaotic. Turbulent flow is more effective at removing heat from a surface due to the rapid movement of the fluid molecules compared to laminar flow [16]. Two dimensionless numbers define the location where the transition from laminar to turbulent flow occurs on a surface: Rayleigh number and Reynolds number for natural and forced convection, respectively.

The PV module can be modelled as an inclined plate immersed in a fluid. Estimation of the convective heat transfer coefficient highly depends on the fluid flow characteristics around the surface; in turn, it depends on the positioning of the surface with respect to the direction of convective fluid motion. In natural convection, it is possible to identify two primary fluid flow behaviour on the top and bottom surfaces: laminar and turbulent. The flow behaviour interchange between the top and bottom surfaces concerning the direction of the buoyancy force, which depends on the relative temperature of the surface compared to the fluid temperature. In forced convection, wind direction will be the deciding factor in the calculation of the rate of heat transferred from the PV module. Three distinct flow patterns are observed for an inclined plate in an external fluid flow. When the wind blows edge-wise to the module, a parallel flow region develops. Head-on and wake regions form when the wind blows towards the front or back surface of the module. The flow over the surface which wind hits first is called the head-on region. And the wake region forms on the behind surface. In each of these regions for natural and forced convection, a different formulation is used for Nusselt Number. The Nusselt number expressions used in the model are listed in Table 1 [10].

In the heat transfer analyses, usually, one of the two convection types dominates, and the other is neglected. However, in this study, both of them are calculated, then their contribution is combined, since, in outdoor testing, wind speed may vary significantly with time.

2.1.3. Radiative heat transfer

Radiative heat transfer can be defined as the heat transferred by the emission of electromagnetic waves from a surface. Stefan-Boltzmann's law of blackbody radiation governs the process. In the PV modules mounted on a roof, radiation exchange occurs with the sky, earth, and roof. The critical action in the calculation of radiative heat loss is the correct estimation of the temperature of these surfaces. Sky temperature can be calculated using a correlation given in Refs. [17]. Earth and roof temperatures depend on the state of the land surrounding the PV module and materials used in the roof covering, respectively. They are more complicated to estimate. In this study, both are taken to be equal to ambient temperature.

2.2. Electrical power output

PV performance analyses require an accurate estimation of electrical power produced by the module. The power yield of PV cells, especially Si-based cells, strongly depends on the cell temperature. Therefore, the correlation [18] employed for the electrical power output estimation in this model includes module temperature as follows:

$$W_{electrical} = Q_{solar}(\tau\alpha)\eta_{ref}\left[1 - \beta_{ref}(T_m - T_\infty)\right] \tag{10}$$

where $(\tau\alpha)$ is the transmittance-absorbance product, which is a parameter dependent on time and date. T_{ref}, η_{ref}, and β_{ref} are reference temperature, efficiency, and temperature coefficient of the module, respectively.

2.3. Heat capacity of PV module

The heat capacity of a PV module can be approximated to some degree considering layers of the module. It is possible to calculate an average using specific heat capacity values of each layer with the following formulation [17]:

$$C_m = \sum_n d_n \rho_n c_{p_n} \tag{11}$$

Table 1
Nusselt Number expressions for different flow regimes.

Natural Convection				
Flow Regime		Nusselt Number Expression	Eq.	Ref.
Laminar		$\overline{Nu_L} = 0.68 + \frac{0.67\,Ra_L^{1/4}}{[1+(0.492/Pr)^{9/16}]^{4/9}}$	(3)[a]	[16]
Turbulent	$T_m < T_\infty$, BS[b] and $T_m > T_\infty$, TS[c]	$\overline{Nu_L} = 0.13\left[\left(\frac{Ra_L}{\cos\theta}\right)^{1/3} - Ra_c^{1/3}\right] + 0.56(\cos\theta\cdot Ra_c)^{1/4}$	(4)[d]	[20]
	$T_m < T_\infty$, TS and $T_m > T_\infty$, BS	$\overline{Nu_L} = \left\{0.85 + \frac{0.387\,Ra_L^{1/6}}{[1+(0.492/Pr)^{9/16}]^{8/27}}\right\}^2$	(5)	[16]
Forced Convection				
Flow Regime		*Nusselt Number Expression*	*Eq.*	*Ref.*
Parallel	Laminar	$\overline{Nu_L} = 0.664\,Re_L^{1/2}Pr^{1/3}$	(6)[e]	[16]
	Turbulent	$\overline{Nu_L} = (0.037\,Re_L^{4/5} - 871)Pr^{1/3}$	(7)	[16]
Head-on		$\overline{Nu_L} = 0.931\frac{(v_\infty \upsilon L)^{1/2}\rho c_p}{kPr^{2/3}}$	(8)[f]	[21]
Wake		$\overline{Nu_L} = 0.191\,Re_L^{2/3}Pr^{1/3}$	(9)	[16]

[a] $\overline{Nu_L}$ is average Nusselt Number, Ra_L is Rayleigh Number and L is the characteristic length for the corresponding surface and Pr is Prandl Number of the fluid.
[b] BS is the bottom surface of the module.
[c] TS is the top surface of the module.
[d] θ is the angle between the module and the vertical and Ra_c critical Rayleigh Number for the specific.θ.
[e] Re_L is Reynold Number for the corresponding surface.
[f] v_∞, υ, ρ, c_p and k are the speed, kinematic viscosity, density, specific heat capacity and thermal conductivity of the fluid, respectively.

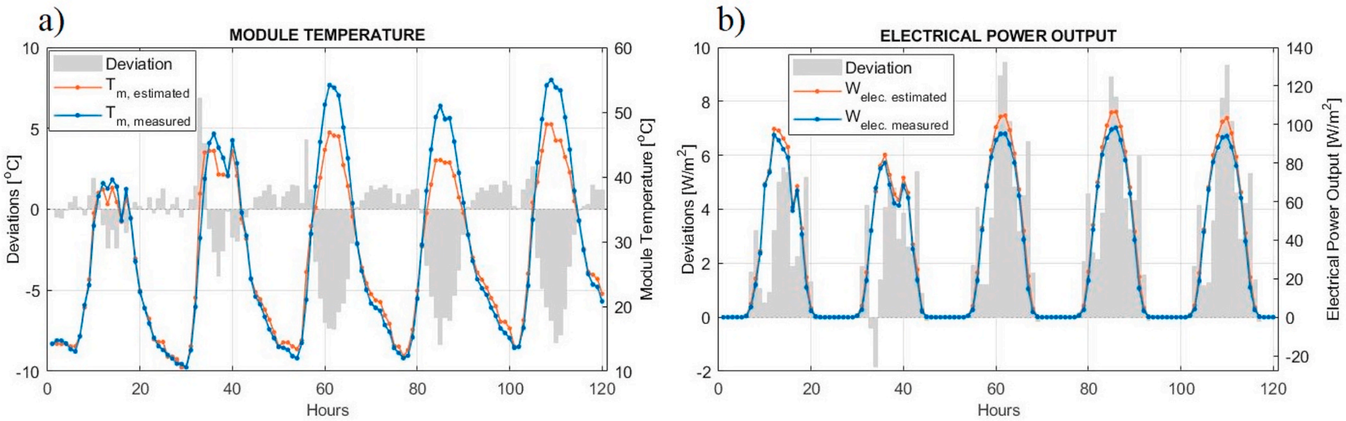

Fig. 1. (a) Measured (blue) and estimated (red) module temperature, and deviations, (b) measured (blue) and estimated (red) electrical output, and deviations. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

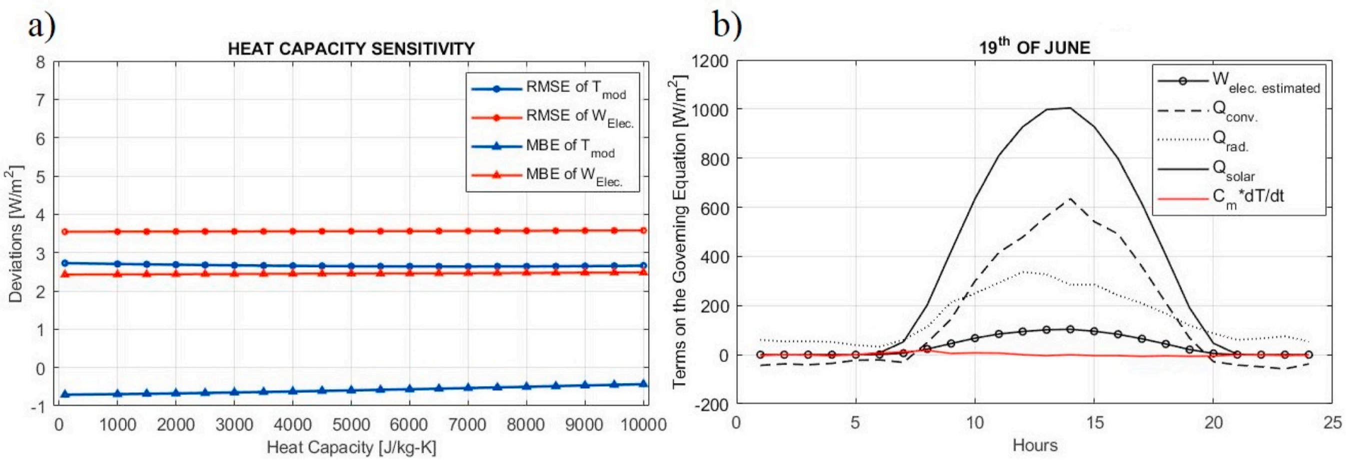

Fig. 2. (a) Heat capacity sensitivity of thermal model performance, (b) power share of each term in the governing equation.

where d, ρ, and c_p are thickness density and specific heat capacity of the nth layer in the PV module, respectively. The possible problem in this approach is that most of the time, the thermophysical properties of these layers are not available because of commercial reasons. Therefore, the effect of heat capacity on the performance of the thermal model should be examined.

2.4. Solution methodology and verification of the model

The solution of the governing equation is realized numerically with implicit Euler's method. The time step for the analysis is taken as 1 h. GÜNAM's Outdoor Testing Facility Research group provided the data for verification of the results [19]. The PV module temperature and electrical power outputs are recorded from a south-facing poly-c-Si panel with the efficiency, temperature coefficient, and inclination angle of 12.7%, 0.45% K^{-1}, and 32°, respectively. Irradiation, ambient and dew point temperature, wind direction, and speed are measured hourly in the weather station of GÜNAM Outdoor Testing Facility. The performance of the proposed model is assessed by statistical deviations of mean absolute error (MAE), root mean square error (RMSE) and mean bias error (MBE).

3. Results

The results of the thermal model are verified with the data measured during a five-day interval from June. The first two days are partially cloudy, and the following three are clear days. The module temperature predictions and measured values are depicted in Fig. 1(a). The thermal model performs better at cloudy days with smaller deviations. The most substantial deviations are observed at the noon hours of clear days. Daytime MAE of module temperature estimations is 2.61 °C. During the night-time, the thermal model predicts module temperatures very accurately with MAE of 0.90 °C. The model tends to underestimate module temperatures; MBE is −1.64 °C during the day. Electrical power output estimations for the PV model can be seen in Fig. 1(b). The results show that yield prediction is better at cloudy days as a result of low module temperature deviations. The model overestimates the electrical power output with MAE and MBE of 3.43 and 3.35 W/m^2, respectively.

Sensitivity analysis for the heat capacity of the PV module is depicted in Fig. 2(a). The calculated value for the heat capacity of the module using equation (4) and the data from [22] is 5723 J/kg-K. The thermal model has been run with the module heat capacity as a variable parameter using C_m values spanning 10^2–10^4 J/kg-K for June. Statistical deviations of the module temperature and electric power output estimations are considered as the deciding factor for the effect of the C_m value choice on the model performance. As can be seen in Fig. 2(a), the RMSE and MBE of T_m and $W_{electrical}$ change very little. Only a slight increase in MBE of T_m estimations with increasing C_m is observed. Therefore, it can be said that the thermal model performance is insensitive to change in heat capacity value. Fig. 2(b) can demonstrate the reason why such substantial changes in heat capacity value do not affect the overall results. In the graph, the numerical values of the terms in the governing equation are depicted for the 19th of June. The red line representing the time-dependent term, which includes heat capacity, has the smallest value. Compared to other terms like the rate of heat loss and electrical power, which are sharing most of the incoming solar energy, the effect of the time-dependent term becomes insignificant.

4. Conclusion

In this study, a transient thermal model for the estimation of the PV module temperature is described. Thermal model performance is verified with data from Outdoor Testing Facility of GÜNAM. The results for a 5-day interval are depicted. The model performed well with MAE and MBE of 2.61 and −1.64 °C for daytime module temperature estimations and MAE and MBE of 3.45 and 3.35 W/m^2 for electrical power output estimations.

Accurate estimation of heat capacity value is critical in transient thermal analyses. Due to the absence of experimental studies concerning PV cell/module heat capacity, a sensitivity analysis performed to determine the effect of selected heat capacity value on the thermal model performance with hourly input data. After the analyses executed using heat capacity values in the interval 10^2 and 10^4 J/kg-K, it is found that RMSE and MBE of module temperature estimations and electrical power output estimations do not vary significantly. The effect of heat capacity value, in turn, time-dependent term in the governing equation on the overall results of the presented thermal model is negligible. Therefore, it can be concluded that for hourly performance analyses of PV modules in semi-arid climate conditions, transient analysis is not needed; steady-state thermal modelling would give satisfactory results.

CRediT authorship contribution statement

B. Tuncel: Software, Formal analysis, Validation, Visualization, Writing - original draft. **T. Ozden:** Investigation, Data curation. **R. S. Balog:** Conceptualization, Methodology. **B.G. Akinoglu:** Investigation, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank to GUNAM, The Center for Solar Energy Research and Applications, and assistance of Dr. R. Turan.

The authors also acknowledge the support of the Ministry of Development during the construction process of GÜNAM's outdoor testing facility.

References

[1] D.M. Chapin, C.S. Fuller, G.L. Pearson, A new silicon p-n junction photocell for converting solar radiation into electrical power, J. Appl. Phys. 25 (5) (1954) 676–677.
[2] M.A. Green, E.D. Dunlop, D.H. Levi, J. Hohl-Ebinger, M. Yoshita, A.W.Y. Ho-Baillie, Solar cell efficiency tables (version 54), Prog. Photovoltaics Res. Appl. 27 (7) (2019) 565–575.
[3] D.S. Philipps, W. Warmuth, Photovoltaics Report, 2019.
[4] A.A. Jadallah, D.Y. Mahmood, Z.A. Abdulqader, Modeling and simulation of a photovoltaic module in different operating regimes, Acta Phys. Pol. A 128 (2) (2015) 461–464.
[5] Z. Er, Z. Rouabah, I.B. Turna, A modal study on the estimation of daily global solar radiation from sunshine duration, Acta Phys. Pol. A 132 (3) (2017) 1070–1075.
[6] Z. Er, I.B. Turna, Solar radiation evaluation by t statistical analysis method for south and north region of Turkey, Arab. J. Geosci. 11 (11) (2018).
[7] M. Bardhi, G. Grandi, G.M. Tina, Comparison of PV cell temperature estimation by different thermal power exchange calculation methods, Renew. Energy Power Qual. J. 1 (10) (2017) 653–658.
[8] E. Skoplaki, J.A. Palyvos, Operating temperature of photovoltaic modules: a survey of pertinent correlations, Renew. Energy 34 (1) (2009) 23–29.
[9] R.S. Balog, Y. Kuai, G. Uhrhan, A photovoltaic module thermal model using observed insolation and meteorological data to support a long life, highly reliable module-integrated inverter design by predicting expected operating temperature,, in: 2009 IEEE Energy Conversion Congress and Exposition, ECCE *2009*, 2009, pp. 3343–3349.
[10] B. Tuncel, B.G. Akinoglu, T. Ozden, R.S. Balog, Thermal modeling and verification of pv module temperature and energy yield using outdoor measurements for Ankara, Turkey, in: *PVCon 2018* - Int. Conf. Photovolt. Sci. Technol., 2018.
[11] S. Armstrong, W.G. Hurley, A thermal model for photovoltaic panels under varying atmospheric conditions, Appl. Therm. Eng. 30 (11–12) (2010) 1488–1495.
[12] D. Torres Lobera, S. Valkealahti, Dynamic thermal model of solar PV systems under varying climatic conditions, Sol. Energy 93 (2013) 183–194.
[13] L. Migliorini, L. Molinaroli, R. Simonetti, G. Manzolini, Development and experimental validation of a comprehensive thermoelectric dynamic model of photovoltaic modules, Sol. Energy 144 (2017) 489–501.
[14] D. Torres-Lobera, S. Valkealahti, Inclusive dynamic thermal and electric simulation model of solar PV systems under varying atmospheric conditions, Sol. Energy 105 (2014) 632–647.
[15] A. Pfreundt, et al., Post-processing thickness variation of PV module materials and its impact on temperature, mechanical stress and power, in: 36th European Photovoltaic Solar Energy Conference and Exhibition, EUPVSEC), 2019.
[16] T.L. Bergman, A.S. Lavine, F.P. Incropera, D.P. Dewitt, Fundamentals of Heat and Mass Transfer, seventh ed., 2011.
[17] J.A. Duffie, W.A. Beckman, Solar Engineering of Thermal Processes, fourth ed., John Wiley & Sons, Inc., New Jersey, 2013.
[18] E. Skoplaki, J.A. Palyvos, On the temperature dependence of photovoltaic module electrical performance: a review of efficiency/power correlations, Sol. Energy 83 (5) (2009) 614–624.
[19] T. Ozden, B.G. Akinoglu, R. Turan, "Long term outdoor performances of three different on-grid PV arrays in central Anatolia – an extended analysis, Renew. Energy 101 (2017) 182–195.
[20] Tetsu Fujii, Hideaki Imura, Natural-convection heat transfer from a plate with arbitrary inclination, Int. J. Heat Mass Tran. 15 (4) (1972) 755–767, https://doi.org/10.1016/0017-9310(72)90118-4.
[21] Sparrow, Tien, Forced Convection Heat Transfer at an Inclined and Yawed Square Plate—Application to Solar Collectors, J. Heat Tran. 99 (4) (1997), https://doi.org/10.1115/1.3450734.
[22] Jones, Underwood, A thermal model for photovoltaic systems, Sol. Energy 70 (4) (2001) 349–359, https://doi.org/10.1016/S0038-092X(00)00149-3.